

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	B . Nandini	Reg. No.:	192525137
Date:	23-08-2026	Duration:	60 minutes

Code of Conduct

I B. Nandini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.Nandini

Annexure A – Test Case Design Assignment

Agriculture equipment rental and farm resource management platform

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.

Functional Requirements

1. User registration and login.
2. Farmer profile creation and management.
3. Equipment owner registration.
4. Add, update, and delete agricultural equipment.
5. Search equipment based on type, location, availability, and rental price.
6. View equipment details.
7. Check equipment availability.
8. Book/rent agricultural equipment.
9. Select rental start and end dates.
10. Calculate rental cost automatically.
11. Confirm or cancel rental bookings.
12. Equipment owner can accept or reject rental requests.
13. Track current and previous rentals.
14. Manage farm resources such as seeds, fertilizers, tools, and machinery.
15. Update resource quantity.
16. Generate rental and resource-management reports.
17. Send booking and status notifications.
18. Provide feedback/rating for rented equipment.
19. Admin can manage users and equipment.
20. System maintains rental and transaction history.

2. Non-Functional Requirements

1. **Performance:** Search and booking operations should respond quickly.
2. **Security:** User credentials and transaction information must be protected.
3. **Reliability:** The system should prevent double booking of equipment.
4. **Usability:** Interface should be simple and farmer-friendly.
5. **Availability:** The platform should be accessible whenever required.
6. **Scalability:** The system should support increasing numbers of farmers and equipment.
7. **Maintainability:** Software should be easy to update and maintain.
8. **Data Integrity:** Rental and resource information must remain accurate.
9. **Compatibility:** System should work on common browsers and mobile devices.
10. **Privacy:** Personal and payment-related information should not be exposed.

2. Identify the **test scenarios** for the assigned problem statement.

Scenario ID	Test Scenario
TS01	Register a new farmer
TS02	Login with valid credentials
TS03	Login with invalid credentials
TS04	Update farmer profile
TS05	Add agricultural equipment
TS06	Update equipment details
TS07	Delete equipment
TS08	Search available equipment
TS09	Filter equipment by location
TS10	Filter equipment by rental price
TS11	View equipment details
TS12	Check equipment availability
TS13	Book available equipment
TS14	Book already-rented equipment
TS15	Select valid rental dates
TS16	Enter invalid rental dates
TS17	Calculate rental cost
TS18	Confirm booking
TS19	Cancel booking
TS20	Owner accepts rental request
TS21	Owner rejects rental request
TS22	Track rental status
TS23	Add farm resource
TS24	Update resource quantity
TS25	Search farm resources
TS26	Generate resource report
TS27	Submit equipment feedback
TS28	Send booking notification
TS29	Prevent unauthorized access
TS30	Handle server/database failure

3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	Farmer registration	Open registration → enter details → submit	Valid name, email, phone, password	Account should be created successfully	To be recorded	Pass/Fail
TC02	Duplicate registration	Register using existing email	Existing email	System should reject duplicate account	To be recorded	Pass/Fail
TC03	Login with valid credentials	Enter username and password → Login	Valid credentials	User should successfully login	To be recorded	Pass/Fail
TC04	Login with invalid password	Enter incorrect password	Wrong password	Error message should be displayed	To be recorded	Pass/Fail
TC05	Empty login fields	Click Login without entering data	Blank fields	Required-field validation should appear	To be recorded	Pass/Fail
TC06	Update profile	Open profile → modify details → Save	Updated phone/address	Profile should be updated	To be recorded	Pass/Fail
TC07	Add equipment	Owner → Add Equipment → enter details → Save	Tractor, ₹1000/day	Equipment should be added	To be recorded	Pass/Fail
TC08	Invalid equipment price	Enter negative rental price	-₹500	System should reject the value	To be recorded	Pass/Fail
TC09	Update equipment	Select equipment → Edit → Save	Updated price	Equipment details should change	To be recorded	Pass/Fail
TC10	Delete equipment	Select equipment → Delete	Equipment ID	Equipment should be removed after confirmation	To be recorded	Pass/Fail
TC11	Search equipment	Enter equipment type → Search	Tractor	Matching equipment should be displayed	To be recorded	Pass/Fail
TC12	Search unavailable equipment	Search equipment not available	Equipment ID	Unavailable equipment should not be bookable	To be recorded	Pass/Fail
TC13	Filter by location	Select location → Search	Andhra Pradesh	Equipment from selected location should appear	To be recorded	Pass/Fail
TC14	Filter by price	Set price range → Search	₹500–₹1500	Equipment within range should appear	To be recorded	Pass/Fail
TC15	View equipment details	Select equipment	Tractor ID	Name, price, location and availability should display	To be recorded	Pass/Fail

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC16	Valid booking	Select available equipment → choose dates → Book	10/09/2026–12/09/2026	Booking should be created	To be recorded	Pass/Fail
TC17	Booking unavailable equipment	Select equipment already rented	Same dates	Booking should be rejected	To be recorded	Pass/Fail
TC18	Invalid start date	Select past date	01/01/2025	System should reject invalid date	To be recorded	Pass/Fail
TC19	End date before start date	Enter start after end	Start: 15/09, End: 12/09	Booking should be rejected	To be recorded	Pass/Fail
TC20	Same start/end date	Enter same valid date	15/09/2026	System should handle according to rental-day rule	To be recorded	Pass/Fail
TC21	Calculate rental amount	Select equipment and rental period	₹1000 × 3 days	Total should be ₹3000	To be recorded	Pass/Fail
TC22	Confirm booking	Complete booking → Confirm	Valid booking	Booking status should become Confirmed/Pending	To be recorded	Pass/Fail
TC23	Cancel booking	Open booking → Cancel	Booking ID	Booking should be cancelled	To be recorded	Pass/Fail
TC24	Owner accepts request	Owner opens pending request → Accept	Valid rental request	Status should change to Accepted	To be recorded	Pass/Fail
TC25	Owner rejects request	Owner selects Reject	Rental request	Status should change to Rejected	To be recorded	Pass/Fail
TC26	Double booking prevention	Two farmers book same equipment for same period	Same equipment/dates	Only one booking should succeed	To be recorded	Pass/Fail
TC27	Add farm resource	Open resources → Add	Fertilizer, 100 kg	Resource should be added	To be recorded	Pass/Fail
TC28	Update resource quantity	Edit resource quantity	100 → 150 kg	Quantity should update	To be recorded	Pass/Fail
TC29	Negative resource quantity	Enter negative quantity	-20 kg	System should reject input	To be recorded	Pass/Fail
TC30	Resource search	Search resource	Fertilizer	Matching resource should appear	To be recorded	Pass/Fail
TC31	Resource report	Select report → Generate	Monthly resource data	Correct report should be generated	To be recorded	Pass/Fail

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC32	Submit rating	Complete rental → submit rating	5 stars + comment	Feedback should be stored	To be recorded	Pass/Fail
TC33	Invalid rating	Enter rating outside allowed range	0 or 6 stars	System should reject invalid rating	To be recorded	Pass/Fail
TC34	Notification	Confirm booking	Valid booking	User should receive booking notification	To be recorded	Pass/Fail
TC35	Unauthorized access	Farmer tries to access owner/admin page	Farmer account	Access should be denied	To be recorded	Pass/Fail
TC36	Session logout	Click Logout	Logged-in user	Session should terminate	To be recorded	Pass/Fail
TC37	Database failure	Perform booking during database failure	Valid booking	System should display suitable error and avoid corrupt booking	To be recorded	Pass/Fail
TC38	Network interruption	Disconnect network during booking	Valid booking	System should handle failure without duplicate booking	To be recorded	Pass/Fail
TC39	Large equipment search	Search with many records	1000+ equipment records	Results should load within acceptable time	To be recorded	Pass/Fail
TC40	Mobile compatibility	Open system on mobile browser	Android/iOS browser	Interface should display and function correctly	To be recorded	Pass/Fail

4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status.**

Equivalence Partitioning divides input data into valid and invalid groups.

Input	Valid Partition	Invalid Partition
Rental Price	Price > 0	Price ≤ 0
Quantity	Quantity > 0	Quantity ≤ 0
Email	Valid email format	Invalid email format
Password	Meets minimum requirement	Too short/empty
Rental Date	Valid future/allowed date	Past/invalid date
Rating	1–5	Below 1 or above 5
Phone Number	Valid number format	Alphabetic/invalid length
Equipment Type	Existing equipment type	Empty/non-existing type

Boundary Value Analysis tests values at and around limits.

Parameter	Boundary Values	Expected Result
Rating	0, 1, 5, 6	1–5 accepted; 0 and 6 rejected
Quantity	0, 1	Positive quantity accepted
Rental Price	0, 1	Positive price accepted
Rental Period	Minimum allowed period ± 1	Boundary should be correctly handled
Password Length	Minimum-1, Minimum, Minimum+1	Only valid length accepted
Search Result Count	0, 1, Maximum, Maximum+1	System should handle limits correctly

5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.

Testing Techniques

- **Equivalence Partitioning:** Valid/invalid price, quantity and email.
- **Boundary Value Analysis:** Minimum and maximum rating, quantity and rental period.
- **Decision Table:** Booking based on login, availability and valid dates.
- **State Transition:** Available $\rightarrow$ Requested $\rightarrow$ Accepted $\rightarrow$ Rented $\rightarrow$ Returned.

6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Test Coverage

- Functional testing
- Security testing
- Performance testing
- Boundary testing
- Invalid input testing
- Exception handling
- Compatibility testing

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25